

Name of the Student	Narra Sai Mounika
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GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 3

Technical Presentation

COURSE NAME: Deep Learning
SLOT: A

COURSE CODE: MLA0403
Faculty: Dr. Arul Raj A.M

COURSE OUTCOME COVERED

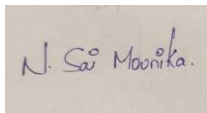
CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35%
Assessment Tool Weightage: 50%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I Narra Sai Mounika / 192425367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

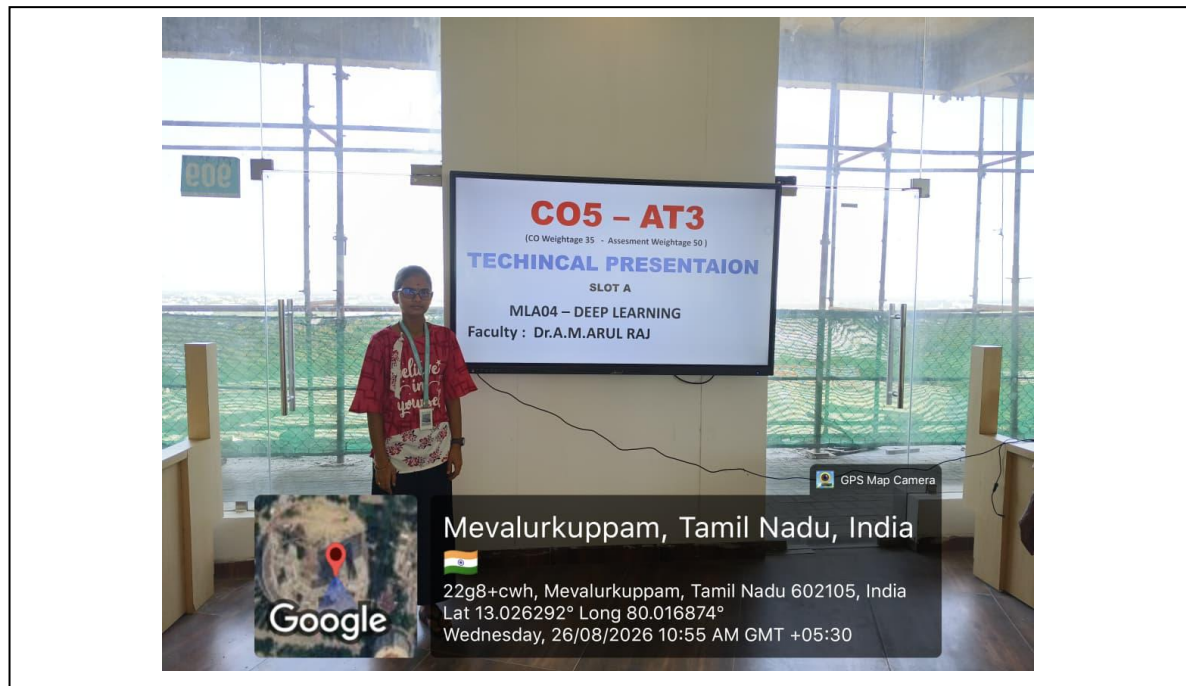
Narra Sai Mounika
Name of the student:

Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (3)	Accuracy of Solution (2)	Clarity (1)	Total (10)
Weightage	20 %	30%	30%	20 %	10 %	100%
Q1						
Total						

Signature of the Faculty:

Total Marks:

PHOTO :



QUESTIONS:

Deep Boltzmann Machines

Prepare a technical presentation on Deep Boltzmann Machines, explaining their multilayer architecture, learning mechanism, interaction among visible and hidden layers, and advantages over conventional Boltzmann Machines. Discuss a practical application and justify why a Deep Boltzmann Machine is appropriate for the selected problem.

PRESENTATION ANSWER:

Introduction

- Probabilistic deep learning model.
- Consists of visible and multiple hidden layers.
- Learns features from unlabeled data.

Architecture

- One visible layer.
- Multiple hidden layers.
- Connections only between adjacent layers.

Layer Interaction

- Visible layer receives input.
- Hidden layers learn features hierarchically.
- Information flows in both directions.

Learning Mechanism

- Pretraining using RBMs.
- Fine-tuning of all layers.
- Uses Gibbs Sampling for learning.

Advantages

- Learns complex patterns.
- Better feature extraction.
- Higher accuracy than traditional Boltzmann Machines.
- Handles large datasets effectively.

Application – Handwritten Digit Recognition

- Dataset: MNIST.
- Learns digit features automatically.
- Classifies handwritten digits accurately.

Why DBM?

- Captures hierarchical features.
- Works well with unlabeled data.
- Improves recognition performance.

Challenges

- High computational cost.
- Slow training process.
- Requires more memory.

Conclusion

- DBM is a deep generative model.
- Uses multiple hidden layers for learning.
- Suitable for complex pattern recognition tasks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand